View Report

R1

(Number of First Attempts: 88)

HW1

Question 1 Difficulty: 1

For a model with single dependent variable against a market benchmark (CAPM beta estimation), assuming that the probability of outlier values in the benchmark is negligible, but making no assumptions on the distribution of dependent variable returns, which of the below estimation methods is most feasible?

➡ Huber regression	67	(62.04 %)
OLS	40	(37.04 %)
WLS	0	(0 %)
GLS	1	(0.93 %)

Average Grade: 0.62 / 1 (62.04 %)

Question 2 Difficulty: 1

Which of the following methodologies is best suited for automated feature selection in a linear regression framework?

Ridge	0	(0 %)
➡ Lasso	108	(100 %)
Huber	0	(0 %)
OLS	0	(0 %)

Average Grade: 1 / 1 (100 %)

Question 3 Difficulty: 1

Can the point of tangency between objective function and constraint region in Ridge regression be on any of the axes (i.e. setting a variable coefficient therefore to 0)?

Average Grade: 0.94 / 1 (93.52 %)

➡ Generally no, except in corner case where solution to corresponding OLS optimization problem is also situated exactly on one of the axes	101	(93.52 %)
No, in all cases	4	(3.7 %)

Yes, due to the angular topology of the constraint region in Ridge	0	(0 %)
Yes, due to the ovoid topology of the constraint region in Ridge	3	(2.78 %)

Question 4 Difficulty: 1

Outlier incidence in a variable will: (choose best answer)

Average Grade: 0.85 / 1 (85.19 %)

Decrease the incidence of Type 2 errors, therefore resulting in significant p values where the p values absent outliers may otherwise be insignificant	0	(0 %)
Increase the incidence of Type 1 errors, therefore resulting in insignificant p values where the p values absent outliers may otherwise be significant	16	(14.81 %)
➡ Increase the incidence of Type 2 errors, therefore resulting in insignificant p values where the p values absent outliers may otherwise be significant	92	(85.19 %)
Decrease the incidence of Type 1 errors, therefore resulting in	0	(0 %)

significant p values where the p values absent outliers may otherwise be insignificant

Question 5 Difficulty: 1

Y intercept value in a Theil-Sen regression is not resistant to outliers, although the slope coefficicient is resistant. Is this statement:

True, because the Y intercept is a function of sample mean, while the slope in Theil-Sen is a function of sample medians	25 (23.15 %)
➡ False, because the Y intercept is a function of sample median, similar to the slope in Theil-Sen which is also a function of sample medians	69 (63.89 %)
True, because the Y intercept is a function of sample median, similar to the slope in Theil-Sen which is also a function of sample medians	11 (10.19 %)
False, because the Y intercept is a function of sample median, while the slope in Theil-Sen which is a function of sample means	3 (2.78 %)

Average Grade: 0.64 / 1 (63.89 %)

Question 6 Difficulty: 1

We are estimating market beta of an instrument traded in a small stock exchange located somewhere in Antartica (fictional country) where domestic market index of instruments on that stock exchange over last 3

years (using daily data) has demonstrated a kurtosis value in excess of 100. Which of following estimation methodologies is the best for this market beta estimation

OLS	0	(0 %)
Ridge	0	(0 %)
Huber	38	(35.19 %)
➡ Theil Sen	70	(64.81 %)

Average Grade: 0.65 / 1 (64.81 %)

Question 7 Difficulty: 1

Theil-Sen is able to demonstrate resilence to arbitrary corruption in up to 29.3% of the data. What does arbitrary mean?

➡ Randomly distributed corruption without any selection bias	48	(44.44 %)
Both randomly distributed as well as predictable coruption in the data	59	(54.63 %)
Only predictable corruption in the data	0	(0 %)
None of the above	1	(0.93 %)

Average Grade: 0.44 / 1 (44.44 %)

Question 8 Difficulty: 1

Which of the following is the best methodology in a situation with high degree of collinearity, if we are limited to one of the options below?

Average Grade: 0.93 / 1 (92.59 %)

Generalized Least Squares (GLS) because of heteroskedasticity	0	(0 %)
➡ Ridge regression, because collinearity often results in extreme numerical value of the estimated coefficients via OLS	100	(92.59 %)
Ridge regression, because	8	(7.41 %)

collinearity often results in numerical value of the estimated coefficients via OLS that is biased towards 0

Generalized Least Squares (GLS) because of homoskedasticity — 0 (0 %)

Question 9 Difficulty: 1

Which of the following methodologies allow for predictable correlation in the error term between different observations?

Weighted least squares, because on diagonal terms of the correlation matrix can vary according to heteroskedasticity — 1 (0.93 %)

➡ Generalized least squares, because off diagonal terms of the correlation matrix of error terms can vary to reflect predictable correlation — 102 (94.44 %)

Average Grade: 0.94 / 1 (94.44 %)

This is already done in basic OLS, which exploits non zero covariance between the dependent and independent variables — 0 (0 %)

All of the above — 5 (4.63 %)

Question 10 Difficulty: 1

Why is normalization / appropriate scaling of variables prior to using the Ridge and Lasso estimators important?

Average Grade: 0.93 / 1 (92.59 %)

Only Lasso but not Ridge will maximize absolute value of estimated coefficients without any awareness of units / scale of independent variables. Variables which are naturally reported at a higher unit of value (e.g. in kilometers instead of centimeters) will be exponentially upsized by both estimation frameworks if not rescaled, even though this effect does not have any economic meaning		0	(0 %)
Only Ridge but not Lasso will constrain absolute value of estimated coefficients without any awareness of units / scale of independent variables. Variables which are naturally reported at a higher unit of value (e.g. in kilometers instead of centimeters)		0	(0 %)

will be disproportionately penalized by both estimation frameworks if not rescaled, even though this effect does not have any economic meaning

Both Ridge and Lasso will maximize absolute value of estimated coefficients without any awareness of units / scale of independent variables. Variables which are naturally reported at a higher unit of value (e.g. in kilometers instead of centimeters) will be exponentially upsized by both estimation frameworks if not rescaled, even though this effect does not have any economic meaning

8 (7.41 %)

Both Ridge and Lasso will constrain absolute value of estimated coefficients without any awareness of

100 (92.59 %)

units / scale of independent variables. Variables which are naturally reported at a higher unit of value (e.g. in kilometers instead of centimeters) will be disproportionately penalized by both estimation frameworks if not rescaled relative to other variables, even though this effect does not have any economic meaning

Question 11 Difficulty: 1

For market beta estimation of biotech firms against a broad based market index (e.g. SPY), which of the following is the economic meaning of the beta?

Average Grade: 0.81 / 1 (80.56 %)

- ➡ A measure of systematic, non diversifiable and non idiosyncratic risk in the firm; hence outlier events due to news flow should not be judged as a component of market beta — 87 (80.56 %)
- A measure of idiosyncratic risk in the firm; hence outlier events due to news flow should be judged as a component of market beta — 3 (2.78 %)

A measure of systematic, non diversifiable and non idiosyncratic risk in the firm; hence outlier events due to news flow should be judged as a component of market beta	16 (14.81 %)	
A measure of idiosyncratic risk in the firm; hence outlier events due to news flow should not be judged as a component of market beta	2 (1.85 %)	

Question 12 Difficulty: 1

What are the portfolio implications of a highly unstable / jumpy rolling market beta that is prone to outlier values?

Unnecessary portfolio turnover due to overreaction of computed market beta to outliers in recent history	0 (0 %)	
Inaccurate hedging leading to disproportionate and unintentional assumption of market risk in hedge positions	0 (0 %)	Average Grade: 1 / 1 (100 %)
High (true) net exposure to the market when hedging using beta values that is computed over outliers	0 (0 %)	
➡ All of the options	108 (100 %)	

Question 13 Difficulty: 1

What is underlying economic intuition for underweighing outliers in Huber estimation, and in biasing coefficient estimates towards 0 in Ridge and Lasso estimation?

Instability due to positive feedback from extreme volatility levels in modern industrial economies	30 (27.78 %)
Higher economic costs of Type 2 versus Type 1 errors in finance	55 (50.93 %)
➡ Mean reversion of extreme volatility levels in modern industrial economies	22 (20.37 %)
Equivalent economic costs of Type 2 versus Type 1 errors in finance	1 (0.93 %)

Average Grade: 0.2 / 1 (20.37 %)